

Enriched finite elements and local rescaling for vibrations of axially inhomogeneous Timoshenko beams

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Abstract

This work presents a new enriched finite element method dedicated to the vibrations of axially inhomogeneous Timoshenko beams. This method relies on the “half-hat” partition of unity and on an enrichment by solutions of the Timoshenko system corresponding to simple beams with a homogeneous or an exponentially-varying geometry. Moreover, the efficiency of the enrichment is considerably increased by introducing a new formulation based on a local rescaling of the Timoshenko problem, that accounts for the inhomogeneity of the beam. Validations using analytical solutions and comparisons with the classical high-order polynomial FEM, conducted for several inhomogeneous beams, show the efficiency of this approach in the time-harmonic domain. In particular low error levels are obtained over ranges of frequencies varying from a factor of one to thirty using fixed coarse meshes. Possible extensions to the research of natural frequencies of beams and to simulations of transient wave propagation are highlighted.

Keywords: Wave-based finite element method, Timoshenko beams, vibrations, high frequency

1. Introduction

Test de bibliographie :

2. Notations and available exact solutions to the Timoshenko system

3. Enriched finite elements

4. A new formulation based on a local rescaling

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5. Conclusions

Acknowledgement

Appendix A. Other enriched spaces

References

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